6 August 2026 (Thursday)

BLOOD CIRCULATORY SYSTEM

- Blood circulation → Transporting various nutrients, gases (CO_2, O_2) excretory products, hormones, enzymes, etc. from one part of the body to another part. with the help of blood.

- Two types of circulatory system are found in organism.

(i) Open circulatory system
→ In invertebrates blood leaves the heart reaches to body parts directly
Ex;
- Nematodes
- insects
- Starfish
- goiter, sheep, Ascaris

(ii) Closed circulatory system
→ In vertibrates blood stays enclosed travels through closed vessels
Ex;
- Pces
- Amphibions
- Reptiles
- Aves
- Mammals.

Note
1. A closed circulatory system found in Annelida (Earthworm)
2. A double circulatory system is found in birds and animals.
3. A single circulatory system is found in fish class (Pisces).
4. An incomplete double circulatory system is found in amphibions (frog, Salamander).

Note → Despite being a reptile, crocodile have a double circulatory system.

Human Circulatory System

- Discovered by → Dr. William Harvey in 1628.
- Three main parts → (i) Heart
  (ii) blood
  (iii) Blood vessels.

HEART

① Heart

- origin → From mesoderm
- study → Cardiology
- Doctor → Cardiologist
- location → On the left side of the chest.
- Muscles → Cardiac muscles
- Colour → Radish-pink
- Weight → 250g - 300g
  - (females)
  - (male)
- Shape → Conical / like a child's closed fist.
- Function → To pump blood.

→ One cardiac cycle duration → 0.8 sec
→ Two distinct phase in one cardiac cycle
(i) Contraction → Systole
(ii) Relaxation → Diastole

→ Blood pumped in one cardiac cycle → 70 ml
→ Blood pumped in one minute → Approximately 5l cardiac output

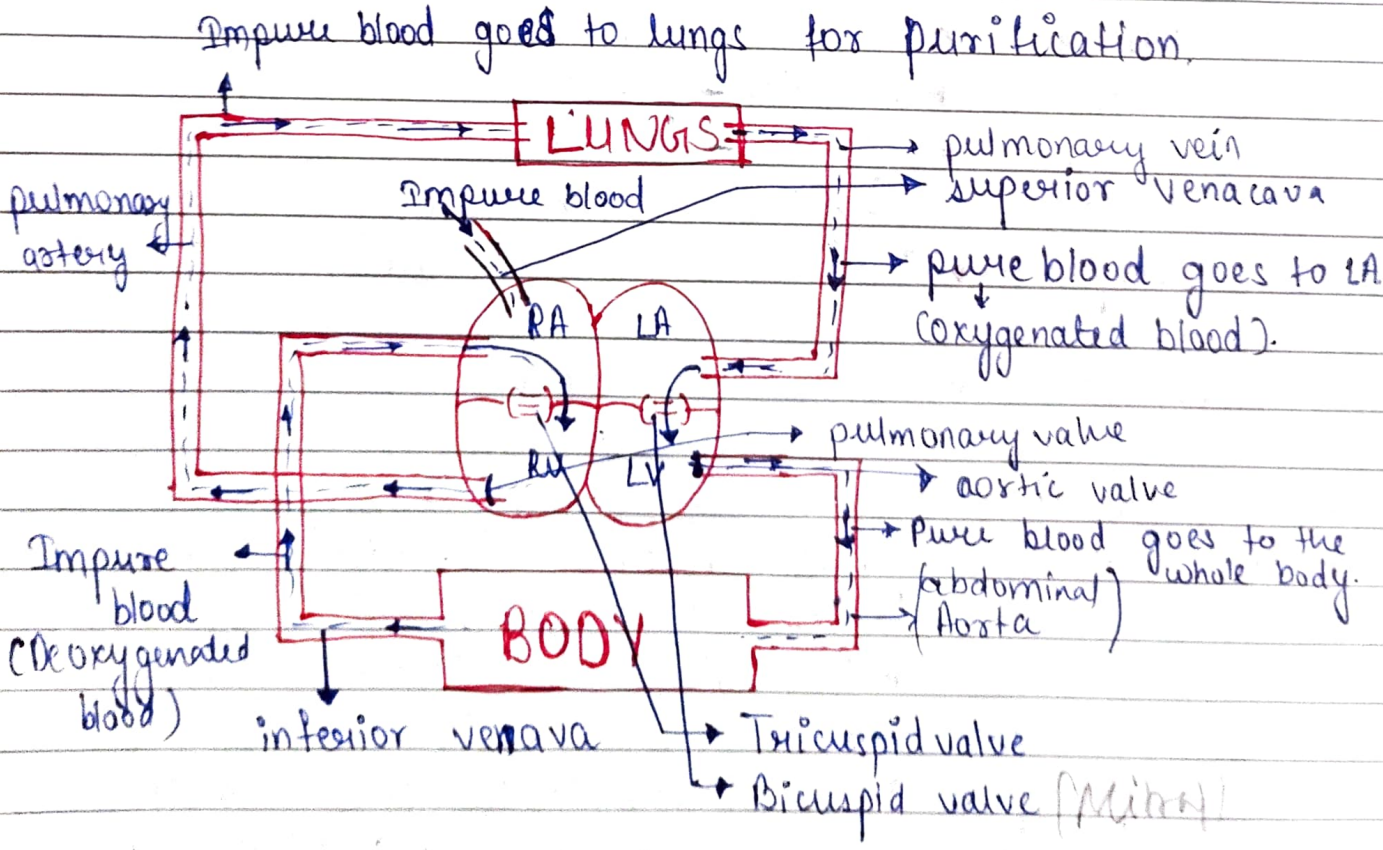
(*) Heart beat

→ Measurment with a stethoscope.
→ At birth → 140-160 times / minute
→ From 1 month to 1 year → 100 times / per minute.
→ In a normal adult → 72 time per minute

→ Lowest → In blue whale (largest mammals).
↳ 25 times per minute.

→ Highest → In shrew (smallest mammal)
↳ 600-800 times per minute.

→ Increase in heart rate → Tachycardia
→ Decrease in heart rate → Bradycardia



7 August 2026 (Friday)

- In the right side → Deoxygenated blood (impure)
- In the left side → Oxygenated (pure) blood
- Purification of blood → in the lungs.
- There are 4 valves in our heart

Valve

- number → 4
- function → to prevent the backward flow of blood
- tests → 2D-ECo, 3D-ECO

→ The closing of the valves produced the Lub-Dub sound.

→ This can be heard using a stethoscope.

→ Pulmonary valve and aortic valve → Semilunar valves.

→ Blood supply to the heart → By coronary artery

→ Obstruction of blood in the coronary → Heart attack

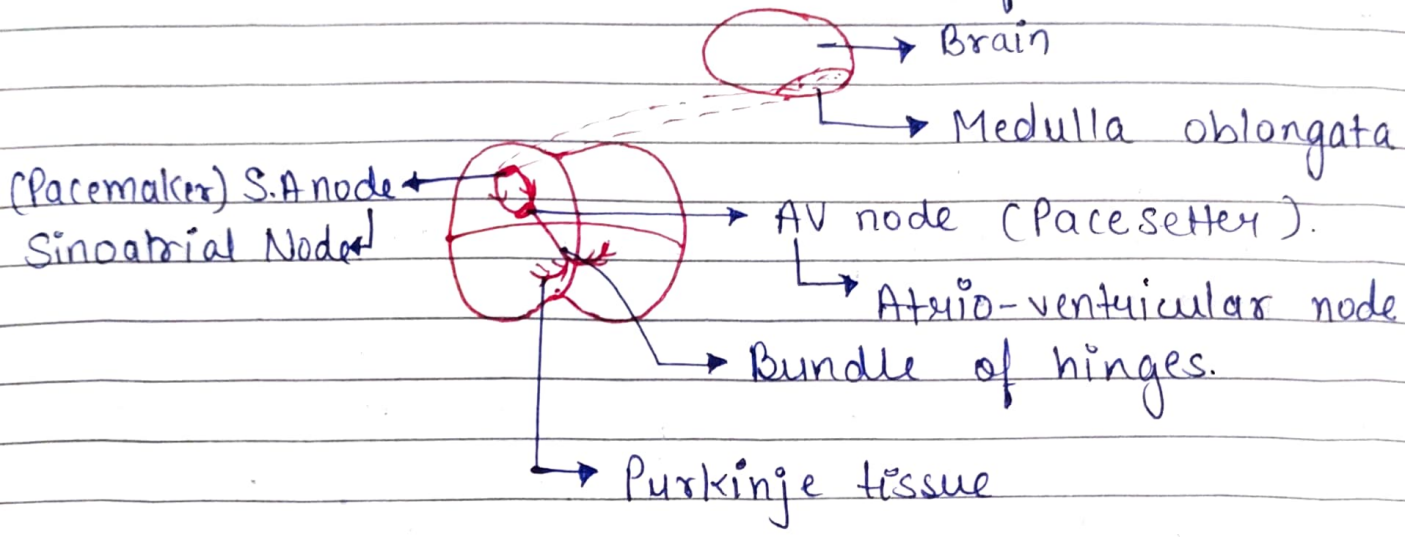
Heart Transplantation

- First in the world → In south Africa
  - → 3 Dec 1967
  - → Dr. Christian Barnar
- First in India → AIIMS (new Delhi).
  - → 3 Aug 1994
  - → Dr. P. Venugopal.

→ Cardiac centres are found in the medulla oblongata in the brain.

→ Pacemaker and pacesetter located in the right atrium.

→ Pacemaker called the 'heart of heart.'



→ Cardiac conduction pathway → S.A. node → A.V node → Bundle of hinges → Purkinje fibres

→ Interruption in the flow of information in the heart → Cardiac Arrest.

→ Examination of heart → ECG (Electrocardiogram)
↳ Discover → Einthoven
first recover by → Waller.

Living being.	No. of chambers in heart.
Birds, mammals, crocodile	4
Fish	2
Amphibian, frog, Salamander	3½
Reptile (snake, lizard) →	2- Atria 1 - incompletely divided ventricle

Cockroach ⟶ 13 chambers

Note → Octopus, Squid have 3 chambered heart.
→ Earthworm have 5 pair of heart like structure.

8 August 2026 (Saturday)

BLOOD

→ study → Hematology
→ Process of formation → Hemopoiesis
→ Destruction → Hemolysis
→ Formation → In red bone marrow
→ In embryonic stage → In liver and spleen
→ Quantity → 5-6 litre or 7-9% of body weight
→ pH value → 7.4
→ Nature → Slightly alkaline
→ Colour → Red

Composition of blood

- Liquid Part
  - Plasma
    - 90-92% (water)
    - 7% (Protein)
    - 1% (other substance)
      - Gases - O_2, CO_2
      - Ions - Na^+, K^+, Cl^-
- Solid Part
  - Blood cells
    - RBC
    - WBC
    - Platelates

Liquid part

PLASMA

⊙ Protein (in blood)

(i°) Albumin
→ Function → Regulates blood pressure.
Transport hormones and medicines.

(ii°) Globulin
→ function → Steenthens immunity.

(iii°) Fibrinogen
→ Function → Helps in blood clotting.

(iv) Hemoglobin
→ Function → Transport oxygen.

→ Plasma is the pale yellow coloured fluid.
→ Hormones, enzymes and anticoagulant substance are found in Plasma. → Heparin

- Functions of plasma.
→ Regulation of blood pH levels.
→ Regulation of osmotic pressure in side blood vessels.
→ Protection against disease and infections.
→ Supporting nutrition and excretion.

Solid pg

Page No.: 83
Date: / /

① RBC (Red Blood cells)

→ Scientific name → Erythrocytes
→ Formation → In red Bone marrow
→ In embryonic stage → In the liver
→ Process of formation → Erythropoiesis
→ Life span → 120 days
→ Deposited in the spleen after destruction
  ↳ Graveyard of RBC
  ↳ Blood Band.

10 August 2026 (Monday)

→ Normal count
  ↳ 50-55 लाख /mm3 → In males
  ↳ 40-45 लाख /mm3 → In females

→ Abnormal decrease in count → Erythrocytopenia
→ Abnormal increase in count → Polycythemia.

→ Decrease in count occurs in Malaria and anemia disease.
→ Increase in count occurs in or during hard physical labour and high altitudes.

→ Size → 7.0 to 8.0 micrometres (um)
→ Increase in size → pernicious Anemia.
→ Shape → biconcave.
→ Change in shape → Sickle cell Anemia
  ↳ Sickle/crescent shape RBC.

Note → The nucleus is not found in the RBCs of mammals.
Exceptions → Camel, Llama, Giraffe.

→ Golgibody and Mitrochondria are also not found in RBC.
→ Largest RBC → Amphiuma
→ Smallest RBC → Musk deer.

HAEMOGLOBIN

↳ It is a structure made of porphyrin ring.
→ It contains iron in the Fe^{2+} state.

Func^{n} → (i) To provide colour to the blood.
(ii) To transport gases

→ It is also called blood pigment and respiratory pigment.

→ $Hb + O_2$ → Oxyhaemoglobin
$Hb + CO_2$ → Carbaminohaemoglobin.

→ 1 gm of haemoglobin transports 1.34 ml of O_2.
→ Amount of haemoglobin → 12-14 gm/ml → females
→ 14-16 gm/ml → males.

② WBC → WHITE BLOOD CELLS :

→ Scientific name → Leukocytes
formation → In the bone marrow

T-lymphocytes and B-lymphocytes develops and matures in the lymphonodes, spleen and thymus gland.

→ Lifespan → Generally 10 days
→ Size → 12 - 20 micron
→ Normal count → 7000 - 11000 / mm^3

11 August 2026 (Tuesday)
→ Abnormal decrease in count → Leukocytopenia
→ Abnormal increase in count → leukemia.
→ Func^n → To develop immunity.
→ They are called Policemen / guards of body.

→ They are generally of 2 types
(i) granulocytes
(ii) Agranulocytes

WBC
- Granulocytes
  - Basophil
  - Eosinophils
  - Neutrophil
- Agranulocytes
  - Monocytes
  - Lymphocytes
    - B-Lymphocytes
    - T-Lymphocytes.

Function
① Basophil
→ Lowest in number → around 1%.
Count increase during allergies.
Produces histamine and serotonin during allergic reactions.

② Eosinophil
→ It is also known as acidophil.
acidophils

→ Count increase in skin disease.

③ Neutrophil
→ Most abundant in number → 60%
→ First to attack bacteria
→ Multilobed Nucleus
→ Drumstick-like structure near the nucleus - Barbody

Barbody → Helps in sex determination
Present in female
Absent in males

④ Monocytes
→ Largest sized WBC
→ Also called Phagocytes.

⑤ Lymphocytes.
→ Smallest WBC
→ Dense nucleus present
→ It is of two types

T- Lymphocytes
→ Longest lifespan upto 10 years
→ Their count decreases in AIDS disease

B- lymphocytes.
→ They produce antibodies during infection

③ PLATELETS

↳ name → Thrombocytes
→ Formation → In bone marrow
→ Life span → 5-9- days
→ Normal count → 1.5 lakh to 4 lakh.
→ Abnormal decrease in count → Thrombocytopenia.
→ Count decrease in Dengue.
→ Shape (Size) → 2 to 3 μm

Function → Helps in blood clotting upon injury
Secretes thromboplastin.

Order of cell count → RBC > Plt. > WBC
Order of cell size → WBC > Plts > RBC

RBC : WBC ⇒ 600 : 1
Measurement of blood cells → By Hemocytometer
unit of measurement → Unit
↳ 1 unit = 350 ml.

Separation of blood components → By centrifugal method.
Separation of plasma from blood → Plasmapheresis.

Serum = Plasma - Fibrinogen.

In blood bank, blood is stored at 4°C.
In blood bank, blood can be stored for upto 42 days.

Anticoagulants are added to prevent blood from clotting in blood bank.

Anticoagulant agents → (i) Sodium citrate $Na_3C_6O_7H_5$
(ii) Sodium oxalate $Na_2C_2O_4$
(iii) Sodium Dextrose
(iv) EDTA (Mostly used)
↳ Ethylene Diamine Tetra Acetic acid

12 August 2026 (Wednesday)

Note → (i) Colour of octopus blood → Blue
↳ due to presence of hemocynin

(ii) Colour of cockroach blood → White → due to absence of hemoglobin.

(iii) The blood of fish amphibians and reptiles is cold.

→ It take 6-8 minutes for blood to clot.

BLOOD GROUP

↳ Our blood groups are based on the AB system

→ There are 4 types → A, B, AB, O
→ Discovered by ↳ Karl Landsteiner
→ AB blood group discoverd by Decastello & Stur

→ Blood groups are determined based on antigens.

Antigen
- Found on the surface of RBC
- Made of glycoproteins.
- Total Number - 2 → A, B
- They determine the blood group type.

Antibody
- Found in the blood Plasma
- Made of glycoprotein
- Total number - 2 → a, b
- They work opposite to antigens

→ The study of antigen and antibody → Serology.

Blood Group	Antigen	Antibody
A	A	a
B	B	b
AB	A,B	—
O	—	a, b

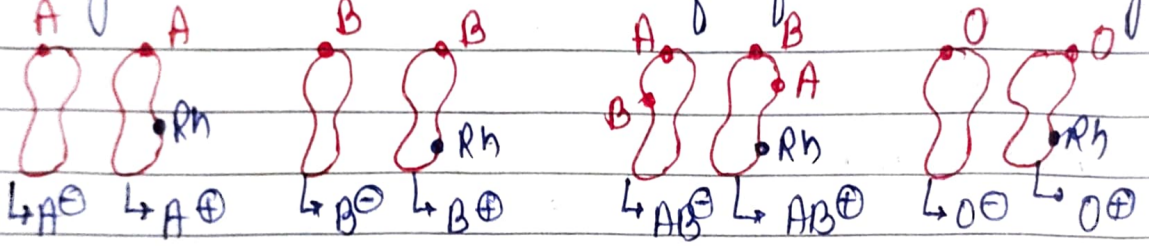
13 August 2026 (Thursday)

Rh - Factor
- Discovered by → Karl Landsteiner & Alexander Wiener.
- Done in Rhesus monkey
- This is also called D - Antigen

→ Those in whom it found → Rh^+
→ Those in whom it not found → Rh^-
→ found on the surface of RBC.

→ When an Rh^+ man marries an Rh^- woman, their 1st child is born normal. Subsequent Rh^+ children die in the womb. This condn is called "Erythroblastosis fetalis".

→ To prevent this, an injection of Rh^0/Anti-D is given at the time of first delivery.

→ 

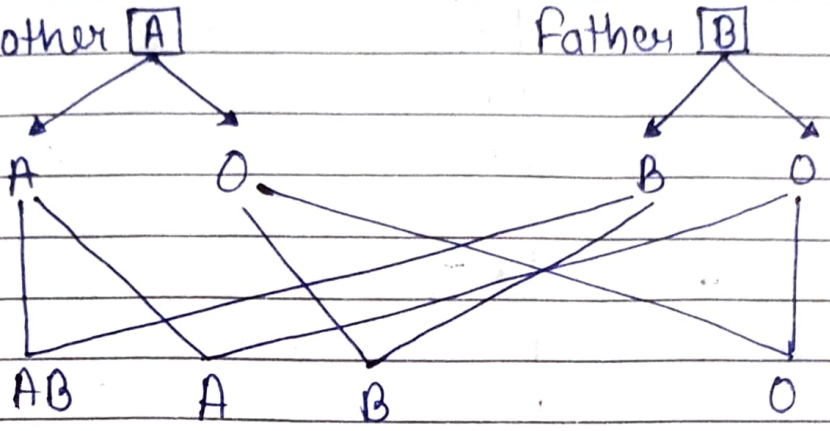
→ Universal donor → $O^\ominus$
→ Universal recipient → $AB^\oplus$
→ Most abundant group → $B^\oplus$ (35%)
→ Least abundant group → $O^\ominus$ (5%)

(*) Genetic determination of blood groups

Q. If the mother blood group is B and the father's blood group is O. What will be the blood group of the offspring?



Q2 If mother is of A and father is of B.



25% → AB [probability]
25% → A
25% → B
25% → O

③ BLOOD VESSELS

→ Study of blood vessels → Angiology
test → Angiography

→ These are of two type ;

(i) Arteries	(ii) Veins
• They carry blood away from the heart	• They bring blood towards the heart
• They contain Pure (Oxygenated) blood.	• They contain impure (deoxygenated) blood.
• Their walls are thick.	• Their walls are thin.
• The value of blood pressure is high	• The value of blood pressure is low.
• These are located in the deeper part of body.	• These are located in the upper part of body.

- Blood pressure is measured in the radial artery
- The largest artery is the Aorta.

- Procedures such as administering glucose, giving injections, taking blood sample and blood transfusions are performed in the veins.
- The largest vein is the "Inferior Venacava". (Postcava).

(*) Blood pressure [B.P];
↳ Pressure exerted by blood on blood vessels.
→ Measurement done by → Sphegmanometer
→ Normal Range → 120 | 80 mm Hg.
→ Increase → Hypertension
→ Decrease → Hypotension